

Introduction to Data Analytics

Code: CSIT366

Module 1

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What is Big Data

- **According to Gartner** – It is **huge-volume**, **fast-velocity**, and **different variety** information assets that demand innovative platform for enhanced insights and **decision making**.
- **A Revolution, authors explain it as** – It is a way to solve all the unsolved problems related to data management and handling, an earlier industry was used to live with such problems. With Big data analytics, you can also unlock hidden patterns and know the 360-degree view of customers and better understand their needs.

Big Data



Big Data is data of very big size which can not be processed with usual tools.

To process such data, we need to be distributed architecture.

Extremely large data sets that may be analysed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.

This data could be structured or unstructured.

Characteristics of Big Data

VOLUME

Data At Rest



Problems related to storage of huge data reliably.
e.g. Storage of Logs of a website, Storage of data by gmail.
FB: 300 PB. 600TB/ day

VELOCITY

Data In Motion



Problems Involving the handling of data coming at fast rate.
e.g. Number of requests being received by Facebook, Youtube streaming, Google Analytics

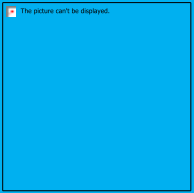
VARIETY

Data in Many Forms

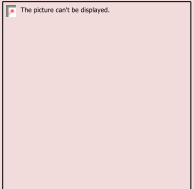


Problems involving complex data structures
e.g. Maps, Social Graphs, Recommendations

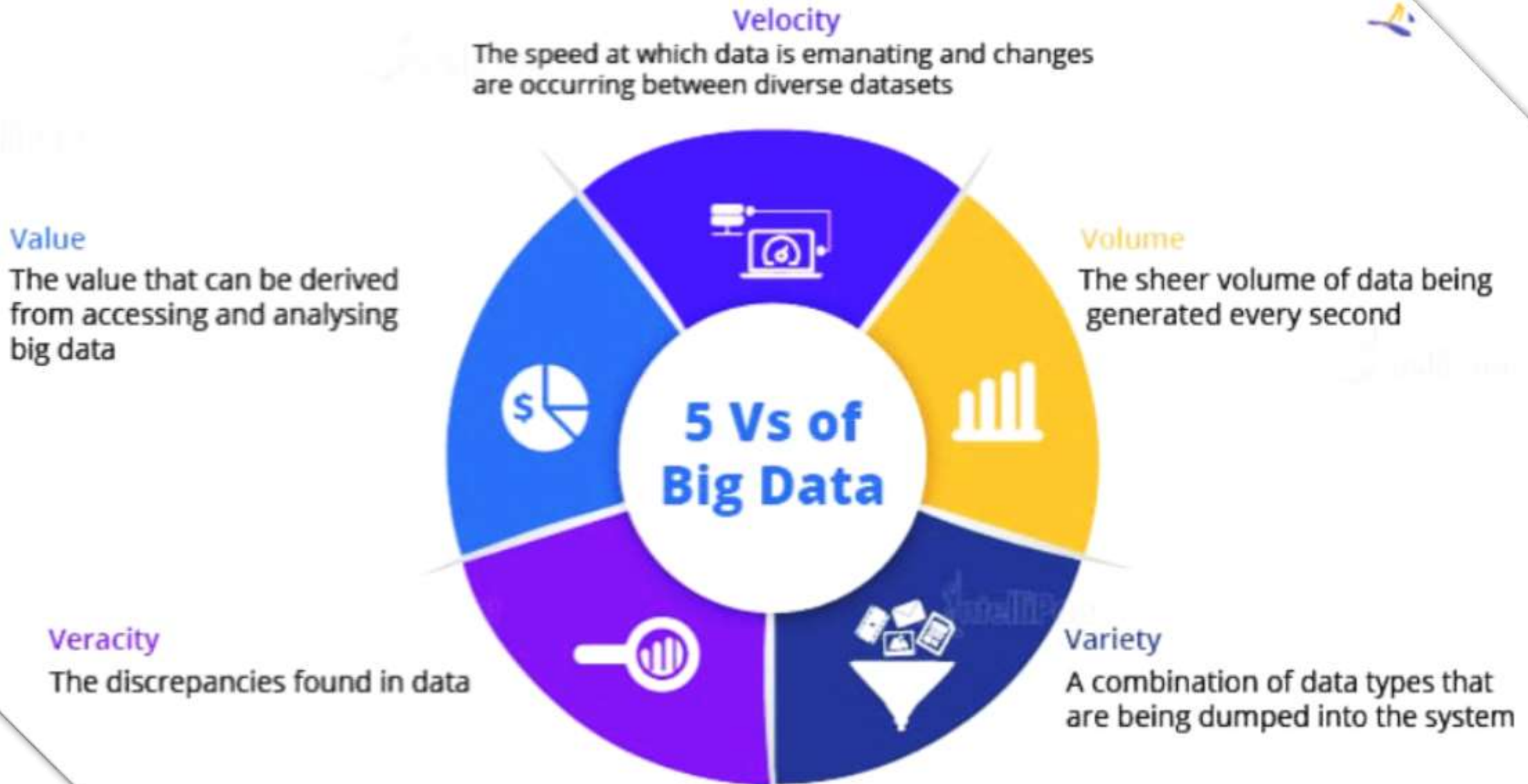
Big Data



Imagine you must find the fastest route on a map since the problem involves enumerating through many possibilities, it is a complex problem even though the map's size would not be too huge.

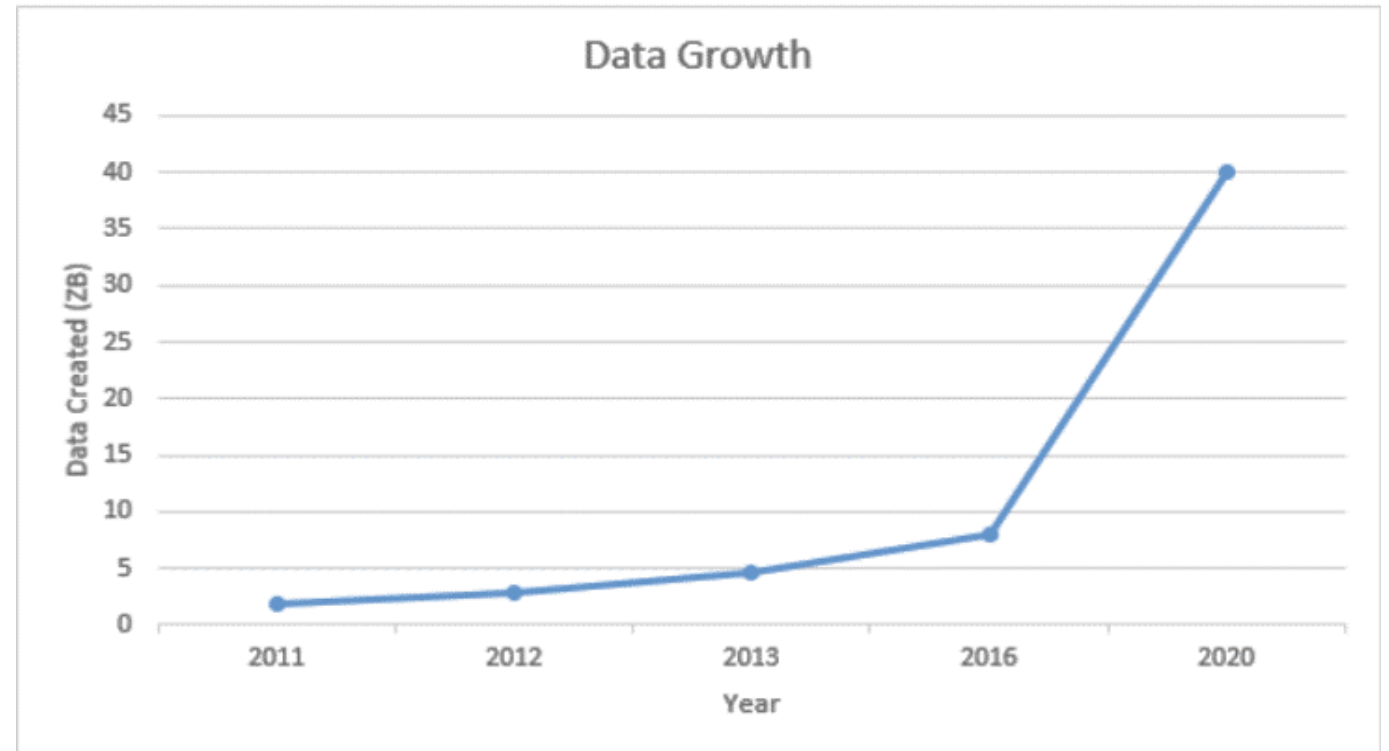


Data could be termed as Big Data if either Volume, Velocity or Variety becomes impossible to handle using traditional tools.



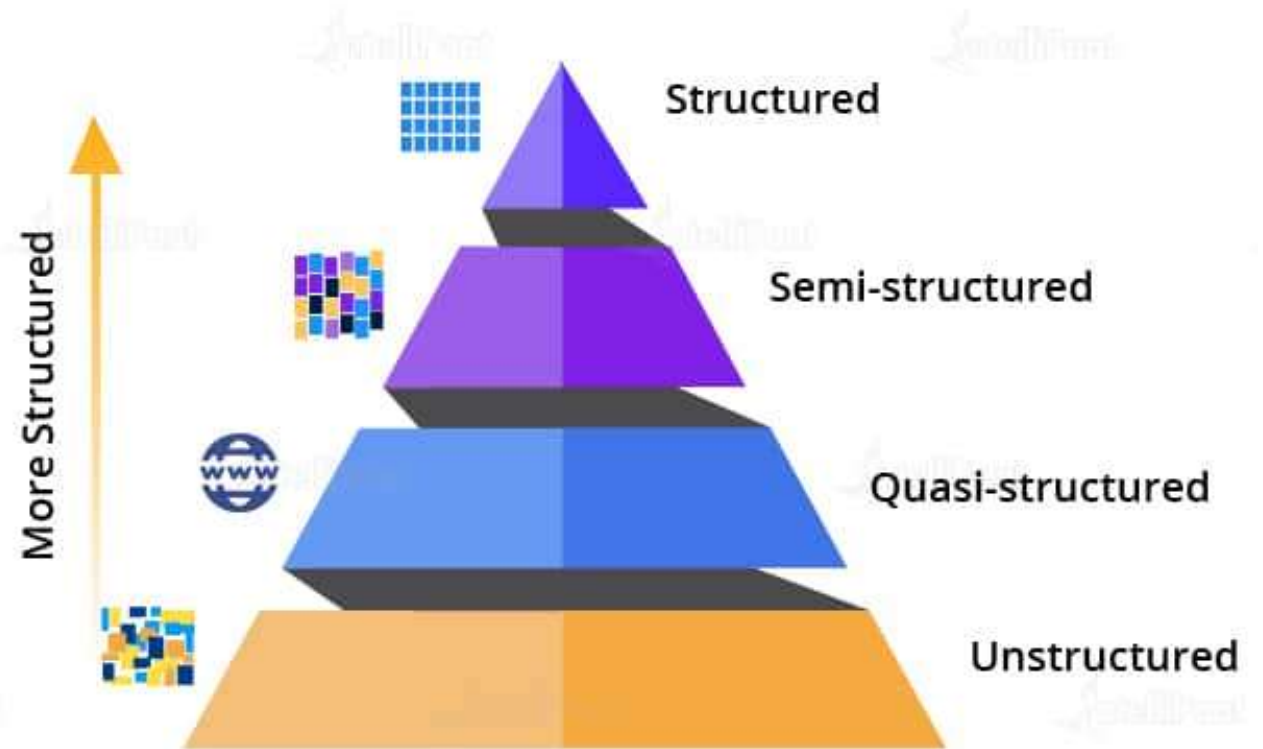
This refers to the data that is tremendously large. As you can see from the image, the volume of data is rising exponentially. In 2016, the data created was only 8 ZB(zeta byte) and it is expected that, by 2020, the data would rise to 40 ZB, which is extremely large.

Big Data: Volume



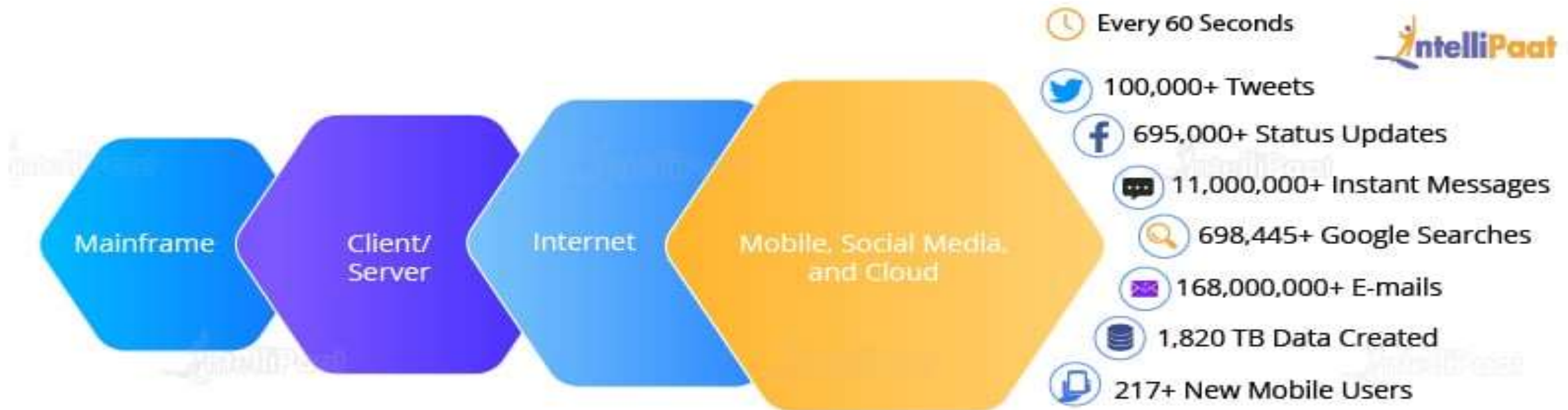
A reason for this rapid growth of data volume is that the data is coming from different sources in various formats. The data is categorized as follows:.

Big Data: Variety



Big Data: **Velocity**

The speed of data accumulation also plays a role in determining whether the data is categorized into big data or normal data.



Big Data: Value

- Deals with a mechanism to bring out the correct meaning out of data.
- First, you need to mine the data, i.e., a process to turn raw data into useful data.
- Then, an analysis is done on the data that you have cleaned or retrieved out of the raw data.
- Then, you need to make sure whatever analysis you have done benefits your business such as in finding out insights, results, etc. which were not possible earlier.



THANK YOU